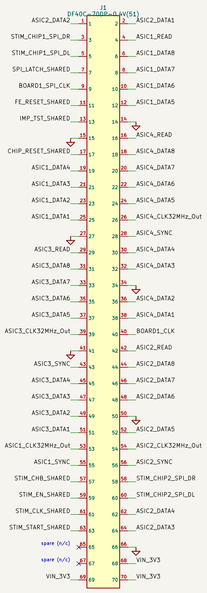
FPGA–LDO PCB V28 VERSION A – ONE ROUTER/LDO BOARD

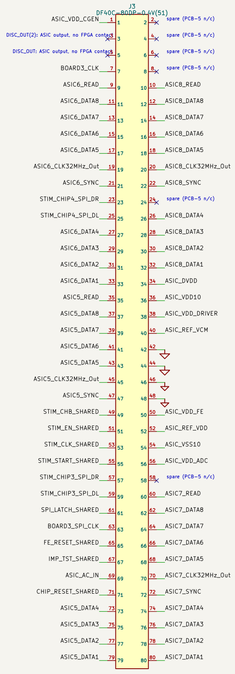
FRONTEND MATING (TOP FACE)



1. Function: Mate the Artix Frontend BL V1 bottom-side receptacles J1/J2 (DF40C-70DS) with two DF40C-70DP plugs on this board's top face, 1.5 mm stack, pin n to pin n. Carries the 108 ASIC signals the two PCB-5 pairs use, VIN_3V3 (J1.68–70, three 0.3 A contacts) and 14 GND contacts.

2. Source / changes: Contact map mirrored pin for pin from the Frontend schematic V43 netlist (2026-11-09_FPGA_Work/01_Frontend_BL_V1_Schematic_V43_J1_J2_Remap/V43_J1_J2_pin_map.csv), which equals remap V3 (2026-11-09_FPGA_Work/03_...). Claude design (remap): J2 keeps the J2-to-J3 order straight (ordered L1/L2 buses): J1 outer-row signal pins carry the left half and inner-row pins the right half of the planned L5 bus to J4. The Frontend re-pinned its FPGA balls to suit (all 8 ASIC groups in their clock bank). Changes from V28: 125 contact assignments: STIM_EN/START/CHB/CLK move from J2 to J1 (J1.59/63/57/61); the 8 unused far-board clock/SPI outputs no longer reach the connectors; spares J1.65/67 and J2.1/2/28/30/32/36–38/40/42/44/49/65 are no-connect (J2.1/J2.2 must stay unassigned). Unchanged: GND 14/15/27/34/41/50/66 on both; J1.68–70 VIN_3V3 (tied on the Frontend in V43). Hirose DF40 catalog, June 2024 (LCSC copy), URL https://datasheet.lcsc.com/datasheet/pdf/145ab5d527b34eafa7a47375cb561c6a.pdf?productCode=C5640153 . DF40C-70DP CZ692097 / 70DS C294854 (J1C, read 2 Nov 2026). Unverified: two rigid DF40 pairs between two boards (the DF40 drawing, sheet 3 item 4, advises one pair per board pair).

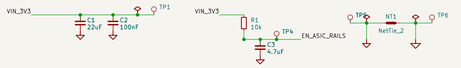
ASIC PORTS – TWO HELENA PCB-5 PAIRS (J3 TOP, J4 BOTTOM, SAME SPOT)



1. Function: Two PCB-5 pairs sandwich this board. J3 (plug, top face) feeds board 3 (near: ASIC5/ASIC6) and board 4 (far, via 1_ lines: ASIC7/ASIC8). J4 (receptacle, bottom face, same XY) feeds board 1 (near: ASIC1/ASIC2) and board 2 (far: ASIC3/ASIC4). Each port also carries the nine ASIC rails; AC_IN, 2 AGND and 2 DGND contacts.

2. Source / changes: ASIC board PCB-5 (Reference/ASICS PCB 1K/PCB – Bottom.zip, PCB.kicad_pcb): J3 DF40C-80DS on B.Cu and J4 DF40C-80DP on F.Cu carry the same net on each pin number, so every contact is bused up the stack: 1_ nets are through-only and feed the second (far) board. Claude pin assignment (as V17): D<k>=DATAk, chip 1 = odd ASIC, chip (2) = even ASIC: the far board shares CLKH, SPI_CLK, SPI_DIN1/R and the stim lines with the near board; DISC_OUT and spares 4/6/8/24/58 open. Unverified: far-board wiring, recording-only boards sharing stim-chip SPI data.

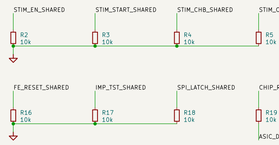
VIN_3V3 ENTRY, DELAYED ENABLE, RETURN BOND



1. Function: C1/C2 hold VIN_3V3 where it arrives on J1.68–70, the three Frontend supply contacts. R1/C3 delay EN_ASIC_RAILS by about 20 ms after VIN_3V3, so all nine ASIC rails start after the frontend 1.5 V bank supply. NT1 is the only AGND–GND bond for all four ASIC boards (the ASIC board must not add its own).

2. Source / changes: Claude design, no manufacturer reference circuit. Delay: tau = R1 x C3 = 47 ms; LT3094 EN/UV turns on at 1.26 V, so t = 47 ms x ln(3.3/2.04) = 23 ms; LT3042 threshold assumed similar (to confirm). Nine EN/UV pin currents x 10k must stay well below 1.26 V. The frontend VCCO_1V5 start time is not measured. NT1 follows the LDO V13 NT100 single-bond idea: PCB-5 J3/J4 carry AGND (42/44) and DGND (46/48) separately and no PCB-5 part joins them; in version B the router carries both returns through J5 and adds no bond (checked in its PCB file). Layout: C901/C1001 return to AGND and the LT3582 output caps to GND, so place NT1 beside U9/U10 to keep converter ripple current out of the ASIC AGND. Unverified: bond location in layout, and whether VIN_3V3 sags at the J1 contacts (0.9 A for three).

ASIC CONTROL SAFE IDLE



1. Function: Hold every FPGA-to-ASIC control line at its safe level while the FPGA is unconfigured: STIM_EN/START/CHB/CLK, FE_RESET, IMP_TEST and SPI_LATCH low; CHIP_RESET high (to ASIC_DVDD), so the ASICs stay in reset until the FPGA releases them. About 150 uA per line when driven to the other level.

2. Source / changes: Claude design from Gerald's firmware (Archive/.../07_FPGA_512_Reference/repository/fpga): top_custom.v section 3b drives CHIP_RESET high for 2^20 cycles (~65 ms) at boot then low (active high); line 264 holds FE_RESET low (amplifier reset released; normal operation"); STIM_* held low; custom_board.lpf line 100 leaves IMP_TEST an input with PULLMODE=DOWN; spi_programmer.v keeps SCLK idle low and pulses LATCH high. Needed because the Frontend ties PUDC_B high (R25 470R), disabling FPGA pull-ups during configuration (AMD UG470, PUDC_B). V23 adds R16–R23 to V22's R2–R9; V28 (Howard, 9 Nov 2026, "Drop R20–R23") removes the four board SPI/recording-clock pulls: while the FPGA configures, CHIP_RESET (pulled high) holds the ASICs in reset and SPI_LATCH (pulled low) blocks any SPI update, so floating clocks have no effect (Claude judgment, not a Gerald rule): it also frees bottom space at the bus taps. READ/SYNC are ASIC outputs and SPI data is only latched by SPI_LATCH, so they get no pull. Unverified: ASIC input leakage and thresholds: reset polarity is from firmware, not an ASIC data sheet.

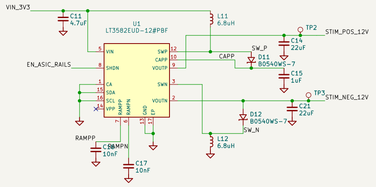
AC_IN DEFAULT



1. Function: R6 holds the analog AC_IN input of all eight ASICs at AGND; TP7 lets a bench source inject a test signal (0–1.5 V nominal) through the shared node.

2. Source / changes: Claude design, The Frontend has no AC_IN source and no connector contact for it (V43), so AC_IN is handled here. V21: in version B these parts sit on the router. so AC_IN needs no J5 contact. Nominal 0–1.5 V from ASIC_REFERENCE.md section 4 (relayed designer answer). Gerald, 9 Nov 2026: "0V is the right state. No need to care about input impedance or anything else."

STIMULATION SUPPLY – VIN_3V3 TO +/–12 V (LT3582-12) THEN +/–10.0 V LDOs



1. Function: LT3582-12 makes STIM_POS_12V (+12 V, boost through D11 to CAPP and its output-disconnect switch to VOUTP) and STIM_NEG_12V (–12 V, single-inductor inverter through D12) from VIN_3V3. as pre-regulators for the stimulation LDOs. Fixed +/–12 V with internal feedback and compensation; starts with EN_ASIC_RAILS (SHDN on above 1.1 V). About 10 mA per output.

2. Source / changes: ADI LT3582/LT3582-5/LT3582-12 data sheet 3582512fb, Figure 13 p.22 (+/–12 V from a single 5 V input), Pin Functions p.8, Choosing Inductors p.18, Capacitor and Diode Selection p.19. https://www.analog.com/media/en/technical-documentation/data-sheets/3582512fb.pdf . Changed from Figure 13: Input 3.3 V instead of 4.5–5.5 V (allowed 2.55–5.5 V; max load at 3.3 V about 40 mA per output from the p.19 equations, need about 10 mA); C1 –> C11 4.7 uF 10 V 0603; C2 –> C21 22 uF 25 V 0805 and C3 –> C14 22 uF 25 V 0805 (about 6–7 uF effective at 12 V, inside the 2.2–10 uF range); C4 –> C15 1 uF 25 V on CAPP; C5/C6 –> C16/C17 10 nF; L1/L2 6.8 uH (Coilcraft XFL2010-682 in the figure) –> Murata DFE201610E-6R8M (Isat to confirm >= 0.42/0.69 A peak per p.18); D1/D2 B0540WS-7 as in the figure; SHDN to EN_ASIC_RAILS instead of an input logic signal; CA/SDA/SCL to GND, VPP open (I2C/OTP unused on the –12 version). V24 replaces the TPS651130 (Howard, 8 Nov 2026; smaller parts) and removes R11–R15, C12, C13, C18–C20, C22. Unverified: JLC stock, ripple after layout, start-up current at the J1 contacts.

STIM SUPPLY SELECT

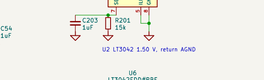
1. Function: JP1/JP2 connect the on-board LT3582-12 outputs to the stimulation LDO inputs. Cut both to feed STIM_PRE_POS/NEG from a bench supply on TP8/TP9 (return: TP6 AGND), as LDO Board A did.

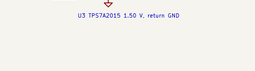
2. Source / changes: Claude design. LDO Board A took INPUT_VDD10/INPUT_VSS10 from outside (Reference/LDO Board A J15/J8). Limits: LT3042 IN <= 20 V, LT3094 IN >= –20 V (ADI data sheets, Absolute Maximum Ratings); keep >= 1.5 V headroom over +/–10.0 V. Leave U1 unfitted, or its outputs open, when external +/–V is used. Unverified: external source noise.

1. Function: U9 (LT3042) makes ASIC_VDD10 = +10.0 V from STIM_PRE_POS; U10 (LT3094) makes ASIC_VSS10 = –10.0 V from STIM_PRE_NEG. Both are referenced to AGND and start with EN_ASIC_RAILS. Each serves the identical eight ASICs (about 8 mA at Gerald's 2 mA per board maximum).

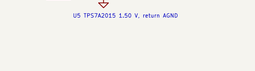
2. Source / changes: U9 inherits Gerald LDO Board A U8 (LT3042, RSET 90k, input from an external INPUT_VDD10). Changed: input is now STIM_PRE_POS (on-board +12 V through JP1, or external); RSET 100k (E96: LDO A 90k). U10 replaces LDO A U9, which floated a positive LT3042 on an external INPUT_VSS10, with a true negative LDO: ADI LT3094 data sheet Rev D (2018), Typical Application p.1 and Pin Functions p.15, https://www.analog.com/media/en/technical-documentation/data-sheets/lt3094.pdf . Changed from p.1: RSET 100k (was 33.2k), CSET 1 uF (was 4.7 uF); PGFB to IN, ILIM to GND, PG and VIOC open, per Pin Functions: EN/UV driven positive (turns on above +1.26 V), V23 (Howard, 8 Nov 2026): +/–10.0 V per Gerald's Nov 2026 pin sheet, so R901/R1001 = 100k (V19–V22 used 90.9k for +/–9.09 V, after LDO A's 90k). About 2 V headroom below the +/–12 V pre-regulators. Confirm the ASIC board VDD10/VSS10 0201 capacitor rating (>= 16 V). V20 output capacitors (Claude change after the 7 Nov review): ceramic capacitance falls with DC bias, so V19's 10 uF 25 V 0603 likely kept only about 3–4 uF at 9.09 V. LT3042 needs >= 4.7 uF and LT3094 >= 10 uF ceramic output capacitance (ADI data sheets, front-page Features). C902 is now 22 uF 25 V 1206; C1002 22 uF 1206 plus C1004 10 uF 25 V 0805. V22: C901/C1001 10 uF 25 V 0805 and C903/C1003 1 uF 25 V 0603 (the SET pins sit at +/–10 V). Unverified: effective capacitance from Samsung DC-bias curves at 10 V.

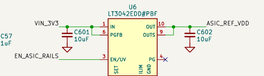
ASIC RAILS – THREE LT3042 (ANALOG / REFERENCE) + FOUR TPS7A20 CELLS FROM VIN_3V3

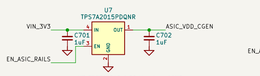


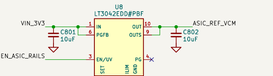












1. Function: Seven regulators make the ASIC rails from VIN_3V3: each rail is shared by all eight ASICs through the two PCB–5 ports (version B: through J5 and the router). LT3042 (U2 VDD_FE, U6 REF_VDD, U8 REF_VCM 0.75 V); Vout = 100 uA x RSET. TPS7A2015 (U3 VDD_ADC, U4 VDD_DRIVER, U5 DVDD, U7 VDD_CGEN): fixed 1.50 V. All start from EN_ASIC_RAILS (TPS7A20 EN high above 0.9 V max, LT3042 EN/UV above about 1.24 V).

2. Source / changes: LT3042 cells inherited from Gerald LDO Board A (Reference/LDO Board A/PCB.kicad_sch): IN, EN/UV and PGFB at the input, ILIM to ground, OUT=OUTS. CIN/COUT 10 uF 10 V 0603, CSET 1 uF, PG open, RSET 15k/7.5k: ADI LT3042 data sheet, Typical Application p.1 and Pin Functions, https://www.analog.com/media/en/technical-documentation/data-sheets/lt3042.pdf. Claude changes: EN/UV to EN_ASIC_RAILS; one regulator per rail for eight ASICs, V24 (Howard, 8 Nov 2026, "find smaller parts"): U3/U4/U5/U7 changed from LT3042 3x3 DFN cells to TI TPS7A2015PDQNR 1x1 mm X2SON (TI SBVS338H Rev H, Jul 2024: Pin Functions p.4 DQN 1 OUT/2 GND/3 EN/4 IN/5 pad: Recommended Operating Conditions p.5 CIN 1 uF, COUT 1–200 uF with >= 0.47 uF effective; Electrical Characteristics p.7 VEN(HI) 0.9 V max: https://www.ti.com/lit/ds/symlink/tps7a20.pdf): their RSET/CSET (R301–R701, C303–C703) are removed and CIN/COUT are 1 uF 10 V 0402. Trade-off: 7 uVrms against 0.8 uVrms (LT3042), accepted by Gerald 9 Nov 2026 ("I think its okay to do that"). VDD_ADC still returns to GND as on LDO Board A. REF_VCM stays LT3042 because 0.75 V is below the TPS7A20 0.8 V minimum. Loads of REF_VDD/REF_VCM unknown.

RAIL CAPACITORS AT THE ASIC PORTS





1. Function: C51 and C53–C57 put 1 uF on six ASIC rails right at the J3/J4 ports, so each has local charge where it enters the PCB–5 stacks: all return to AGND.

2. Source / changes: Claude design (7 Nov review, V20). no manufacturer reference circuit, V28 (Howard, 9 Nov 2026): C52 (VDD_ADC), C58 (VDD10) and C59 (VSS10) removed – no clean layout spot on their rails: VDD_ADC keeps U3's 1 uF C302 beside the pins, VDD10/VSS10 keep 22 uF + 10 uF at U9/U10. V23: C52 (VDD_ADC) moved from GND to AGND to match Gerald's LDO Board A, whose VDD_ADC load decaps C31/C32/C38/C39 go to AGND while its regulator U2 and C8 sit on DGND (as U3/C302 here), and PCB–5 C1/C34 (AGND). In version B the rails reach the router through one J5 contact each and then cross J3/J4, so these parts sit on the router; version A has them too, so A and B stay the same circuit. All are 10 V 0402. Unverified: decoupling already on PCB–5, and the regulators' stability with this extra capacitance at the end of the trace (LT3042/LT3094 are stable with larger output capacitance per their data sheets).

Frontend J1/J2 on top: ASIC ports J3 top, J4 bottom.
3x LT3042 + 4x TPS7A20 ASIC rails; LT3582-12 or external +/–V: LT3042/LT3094 +/–10.0 V.
Schematic only, no layout. Input: VIN_3V3 from Frontend J1.68–70.

Rice – FPGA PCB workspace

Sheet: /
File: FPGA_LDO_V28A.kicad_sch

Title: FPGA–LDO V28 A – one router/LDO board

Size: User	Date: 2026-11-09	Rev: V28
KiCad E.D.A. 10.0.6		Id: 1/1